INSIDE A NORMAL ALGEBRAIC EXTENSION, FINITE SUBEXTENSIONS HAVE FINITE NORMAL CLOSURES, AND NORMALITY SURVIVES FINITE STEPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Item (2) needs M/K normal (not just M'/K), which is exactly what the hypothesis provides; this lemma feeds the Galois-uniqueness machinery for normal closures inside normal towers.

Editorial note. The label lemma-normal-closure-inside-normal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-normal-closure (delivered card L48334: the normal closure of a finite extension is finite and normal), lemma-normally-generated (delivered card L48362 in this batch: adjoining generators whose minimal polynomials split completely gives a normal extension), L/K normal makes every irreducible over K with a root in L split in L

Source reference. Original reference 09DT: sources/stacks-fields/pinned-original-source.tex, source lines 2303–2303. Prerequisite pointer invoked at 2364; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BR3: sources/stacks-fields/pinned-original-source.tex, source lines 2139–2139. Prerequisite pointer invoked at 2372; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be an algebraic normal extension.*

(1) *If $L/M/K$ is a subextension with M/K finite, then there exists a tower $L/M'/M/K$ with M'/K finite and normal.*
(2) *If $L/M'/M/K$ is a tower with M/K normal and M'/M finite, then there exists a tower $L/M''/M'/M/K$ with M''/M finite and M''/K normal.*

Proof. Proof of (1). Let M' be the smallest subextension of L/K containing M which is normal over K. By Lemma 09DT this is the normal closure of M/K and is finite over K.

Proof of (2). Let $\alpha_1, \ldots, \alpha_n \in M'$ generate M' over M. Let $P_1, \ldots, P_n$ be the minimal polynomials of $\alpha_1, \ldots, \alpha_n$ over K. Let $\alpha_{i,j}$ be the roots of P_i in L. Let

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$M'' = M(\alpha_{i,j})$. It follows from Lemma 0BR3 (applied with the set of generators $M \cup \{\alpha_{i,j}\}$) that M'' is normal over K. $\square$